PassFrame tuition

Factors, Multiples & Primes

Year 5 Accelerated — Homework ⏱ 45 min

Worked example — read this through before you start

Worked example: What is the LCM of 4 and 10?

Method & answer: Multiples of 4: 4,8,12,16,20... Multiples of 10: 10,20,30... First common multiple = 20. LCM = 20.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. List the factors of 10.
2. List the factors of 14.
3. List the factors of 25.
4. What are the first 5 multiples of 4?
5. What are the first 5 multiples of 9?
6. Is 23 a prime number?
7. Is 27 a prime number?
8. List the prime numbers between 10 and 20.
9. What is the smallest multiple of 7 that is greater than 50?
10. List the factors of 16.
11. Is 9 a prime number?
12. What are the first 4 multiples of 12?
13. List the factors of 28.
14. Which of these is a multiple of 8: 20, 24, 30?
15. Which of these is a factor of 40: 6, 8, 9?

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Find the HCF of 24 and 40.
2. Find the HCF of 45 and 75.
3. Find the LCM of 8 and 9.
4. Find the LCM of 6 and 15.
5. Write 72 as a product of prime factors.
6. Write 100 as a product of prime factors.
7. List all the common factors of 40 and 60.
8. What is the smallest common multiple of 9 and 12?
9. Is 133 a prime number? (Hint: check 7.)
10. Which of these numbers is prime: 61, 63, 65?
11. Find the HCF of 48 and 72.
12. Find the LCM of 6 and 10.
13. List all factor pairs of 72.
14. Is 126 divisible by 9? (Use the digit-sum rule.)
15. What is the LCM of 3, 5 and 6?

Standard / Advanced — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Find the HCF of the LCM of 4 and 6, and 30.
6. Two joggers run laps of 8 minutes and 12 minutes. If they start together, after how many minutes
11. Three friends visit a museum every 10, 15 and 6 days respectively. If they all visit today, in

will they next be together at the start?

how many days will they all visit together again?

2. List all the factors of the smallest 2-digit prime number.

7. A florist has 36 roses and 54 tulips and wants identical bouquets using all the flowers with none left over. What is the greatest number of bouquets she can make?

12. A number is a multiple of both 9 and 12. What is the smallest such number greater than 100?

3. What is the LCM of the first three prime numbers?

8. A number of pupils, more than 20 and fewer than 40, can be arranged exactly into rows of 6 or rows of 8. How many pupils are there?

13. A field measuring 42 m by 56 m is divided into the largest possible identical square plots with no land left over. What is the side length of each square?

4. Is the HCF of 35 and 49 a prime number?

9. A gardener has ribbon strips of 36 cm, 60 cm and 84 cm. She cuts them into equal pieces, as long as possible, with none wasted. How long is each piece?

14. A baker wants to package 84 croissants and 126 pastries into identical boxes with none left over and the same amount of each per box. What is the greatest number of boxes, and how many of each item per box?

5. Write the smallest even number that has exactly 4 factors.

10. A prime number lies strictly between 72 and 78. What is it?

15. Two lighthouses flash together at midnight, every 18 seconds and every 24 seconds respectively. How many times in the next 300 seconds will they flash together (including the start)?